

Seongkyu CHOI

M.S. Student, Gachon University

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EDUCATION

Gachon University, Seongnam, South Korea (Mar. 2024 – Aug. 2026)

M.S. in Artificial Intelligence

- Department of AI & Software
- Graduate School, Master's Program
- Research Area: Computer Vision, Autonomous Driving, Off-Road, Multi-Task Learning, Edge AI Devices

Hansei University, Gunpo, South Korea (Mar. 2019 - Feb. 2023)

B.S. in Department of International Business

- Department of International Business

Shingu College, Seongnam, South Korea (Mar. 2015 - Feb. 2019)

Associate Degree in Business Administration

- Department of Global Business
- Transferred to Hansei University

RESEARCH INTERESTS

- Semantic segmentation in LiDAR point clouds and utilizing terrain information.
 - Computer Vision for Autonomous Driving and Robotics
 - Multi-Task Perception and Scene Understanding
 - Off-Road Semantic Segmentation under Ambiguous Boundaries and Noisy Supervision
 - Efficient On-Board AI Model Optimization for Edge Devices
 - Multi-Modal Perception using Camera, LiDAR, and Thermal Sensors
 - 3D Occupancy Prediction and Terrain Understanding

WORK EXPERIENCE

Research Experience

- Research Student, Vehicle Intelligence Perception Lab, Gachon University, Korea | Mar. 2024 ~ Aug. 2026
 - Developed computer vision and multi-task perception models for autonomous driving, off-road robots, and edge AI systems.
 - Designed and validated semantic segmentation, traversability recognition and multi-task perception models using camera, LiDAR and edge devices.
 - Implemented on-board AI optimization using Jetson Orin, TensorRT, Docker, ROS2 and evaluated real-time performance using latency, FPS and memory efficiency
 - Validated perception models in real-world robotic and vehicle scenarios, focusing on robustness under complex outdoor conditions.

Work Experience

- Engineer, FNS, Korea | June 2021–Mar. 2023
 - Performed maintenance, troubleshooting and recovery of semiconductor clean-process equipment and provided on-site technical support at customer manufacturing sites.

- Conducted pre-shipment and post-shipment equipment inspections to verify operational stability and quality requirements.
- Analyzed recurring equipment issues and supported preventive actions to improve operational reliability.

PROJECTS

- **Development of Wide Semantic Map Building and Update Technology for Multi-Robot Systems in Unstructured Outdoor Environments**
 - National R&D Project led by Mobiltech | Funded by the Ministry of SMEs and Startups, Korea
 - Project period: Apr. 2025–Mar. 2027 | Participation: Mar. 2024–Aug. 2026
 - ◆ Developed semantic segmentation models integrating RGB and 3D point-cloud information for traversable-area recognition in unstructured outdoor environments.
 - ◆ Evaluated and optimized perception models for wide-area semantic mapping and updates. Contributed to path-similarity recognition and change detection using multi-scan sensor data.
- **Development of Intelligent Battlefield Situational Awareness AI Algorithms**
 - Industry-commissioned Research Project | Hyundai Rotem
 - Project period: Nov. 2024–Sep. 2026 | Participation: Mar. 2024–Aug. 2026
 - ◆ Built RGB/thermal data pipelines covering collection, augmentation, quality review and semi-automatic labeling, expanding the training dataset from approximately 1,500 to 4,000 images.
 - ◆ Deployed a C++ inference pipeline using ONNX and TensorRT, reducing inference latency from 28.29 to 14.18 ms while maintaining mAP50 and mAP50–95. Verified simultaneous processing of two cameras.
- **Development of an Automotive Software Platform for Software-Defined Vehicles (SDV) Integrated with an AI Framework Required for Intelligent Vehicles**
 - IITP-funded National R&D Project led by HL Mando
 - Project period: Apr. 2024–Dec. 2027 | Participation: Apr. 2024–Aug. 2026
 - ◆ Developed and validated an on-board perception system for object detection, lane-line perception and drivable-area segmentation on Jetson AGX Orin.
 - ◆ Improved end-to-end throughput from 3.73 to 24.56 FPS through OpenCV-based visualization optimization, a three-stage asynchronous pipeline and TensorRT FP16 acceleration.
- **Development of a Network for Determining Traversable Regions by Terrain**
 - NIPA-funded Industry-Academic Collaborative Research with Mobiltech and GINT
 - Period: Sep. 2024–Feb. 2025
 - ◆ Developed a camera-only traversability model for agricultural environments using LiDAR knowledge distillation, achieving 0.87 mIoU for semantic segmentation and 0.91 mIoU for traversability classification.
 - ◆ Implemented camera-only segmentation and distance regression on Jetson Orin Nano at approximately 0.1 s/frame.
- **Development of Camera-Based Patrol Cognitive Intelligence for Abnormal Situation Detection**
 - Industry-commissioned Research Project | Neubility
 - Period: Nov. 2024–Mar. 2025
 - ◆ Curated patrol-object data and integrated YOLO detection with vision-language model descriptions using ROS2 and Docker on Jetson Orin Nano.
 - ◆ Applied AWQ quantization to VILA1.5-3B, reducing model size from 10.06 to 1.57 GB and VRAM usage from 8.34 to 5.59 GB.

PUBLICATIONS

International Journal Articles

- **Choi, Seongkyu**; An, Jhonghyun. “Enhancing Group Attention for Off-Road Semantic Segmentation via Transition-Aware Refinement,” IEEE Signal Processing Letters, vol. 33, pp. 2535–2539, 2026.
- **Choi, Seongkyu**; An, Jhonghyun. "A Cross-Scale Decoder with Token Refinement for Off-Road Semantic Segmentation." Applied Sciences, vol. 16, no. 11, Art. no. 5238, 2026.
- Kim, Nahyeong; **Choi, Seongkyu**; Choi, Sun; Lee, Yejun; Cheong, Youngjae; An, Jhonghyun. “GPT-4off: On-Board Traversability Probability Estimation for Off-Road Driving via GPT Knowledge Distillation,” Applied Sciences, vol. 15, no. 4, Art. no. 2130, Feb. 2025.
(Equal contribution.)

RELATED COURSES

- Artificial Intelligence | B0
- Optimization Theory for Machine Learning | B0
- Advanced Topics in AI Autonomous Driving | A0
- Internet and Mobile Computing | A0
- Fashion Big Data Analysis and Application | A+
- Understanding of Advanced Artificial Intelligence | A+
- Probabilistic Robotics | A+
- Intelligent Vehicle Technology | A+

AWARDS

- **President’s Award**, Graduate School, Gachon University | Aug. 2026
- **Outstanding Poster Paper Award**, Korean Society of Next-generation Computing Conference | Apr. 2024
 - Paper: “Real-time Performance Optimization of AI Models Using Edge Computing”

Technical Skills

Programming

Python 
C++ 

Frameworks

PyTorch 
TensorFlow 
OpenMMLab 

Tools

DevOps: Docker, AWS, Jetson Orin, TensorRT
Robotics: ROS2
Tools: Git, GitHub, VScode, Anaconda
OS: macOS, Windows, Linux